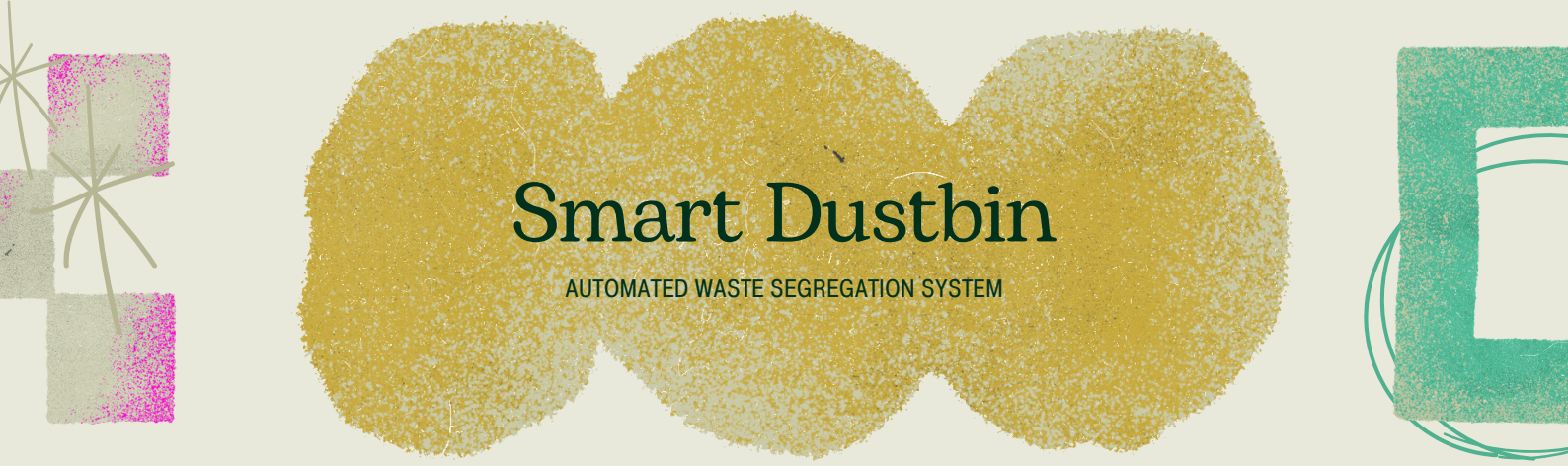




Smart Dustbin

AUTOMATED WASTE SEGREGATION SYSTEM

Aim of the Project

To design and develop an automated smart dustbin system that can intelligently segregate waste into three categories - **wet waste**, **dry waste**, and **metal waste** - using sensors and automation technology for efficient waste management.

Materials Required

- Arduino Uno/Nano Microcontroller
- Moisture Sensor (to detect wet waste)
- Inductive Proximity Sensor (to detect metal waste)
- IR Sensor (to detect the presence of waste)
- Servo Motors (2–3 Nos.) (for rotating the segregation mechanism)
- Ultrasonic Sensor (optional, to detect bin level)
- Three Separate Bins (Green – Wet, Blue – Dry, Grey – Metal)
- Breadboard or PCB
- Jumper Wires / Connecting Wires
- 5V Power Supply or Battery Pack
- Switch and Power Connector
- Dustbin Body and Supporting Frame
- Waste Guide Chute/Funnel
- Acrylic Sheet, Cardboard or Plastic Sheet (for structure)
- Nuts, Bolts and Glue Gun (for assembly)
- LCD Display (Optional) (to display waste type)
- Buzzer (Optional) (for alerts)

Advantages of the Smart Dustbin

The Smart Dustbin offers a multitude of benefits for modern waste management:

- **Automatic Waste Segregation:** Eliminates the need for manual sorting, ensuring efficient separation of wet, dry, and metal waste.
- **Promotes Recycling:** Encourages proper recycling habits by correctly directing waste to designated bins.
- **Reduces Contamination:** Prevents mixing of different waste types, crucial for effective recycling and composting processes.

- **Hygienic Operation:** Features touchless lid operation, minimizing contact and improving sanitation.
- **Environmentally Friendly:** Contributes to a cleaner environment through improved waste management practices.
- **Time and Effort Saving:** Automates the sorting process, saving time and labor.
- **Efficient Disposal:** Facilitates streamlined waste disposal and recycling workflows.
- **Educational Value:** Serves as a tool to raise awareness about environmental conservation and waste management.

Visual Design and Layout

The Smart Dustbin project is designed with a **vibrant and modern aesthetic** to enhance its appeal and functionality. The layout incorporates colorful borders and decorative elements to create an attractive, professional, and visually engaging presentation. Ample space is provided alongside the textual content for **image integration**, allowing for diagrams of the dustbin's structure, sensor placements, or operational flowcharts to be easily incorporated.

Placeholder Image for Dustbin Design

This visual approach ensures that the project documentation is not only informative but also eye-catching, making it ideal for presentations, reports, and educational materials.

Next Steps

Explore potential integration with IoT platforms for real-time waste level monitoring and data analysis. Develop a user-friendly interface for monitoring and controlling the dustbin's operations. Further research into advanced sensor technologies for more accurate waste identification. Consider solar power integration for a more sustainable and autonomous operation.